PROJECT PORTFOLIO · MMXXVI

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A selection of design, simulation, embedded-systems and machine-learning projects spanning my Mechatronics Engineering degree.

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01 THIRD-YEAR PROJECT · INDIVIDUAL · 2025

Machine Learning for the Remaining *Useful Life of Lithium-Ion Batteries*

Trained an LSTM network on the Stanford/MIT battery-discharge dataset to forecast capacity degradation and detect the knee-point of sudden decline, achieving a mean absolute percentage error below 0.07%. Built an interactive Gradio interface for hyperparameter tuning with safeguards against data leakage and battery-boundary mixing — work that maps directly onto risk and asset-default modelling.

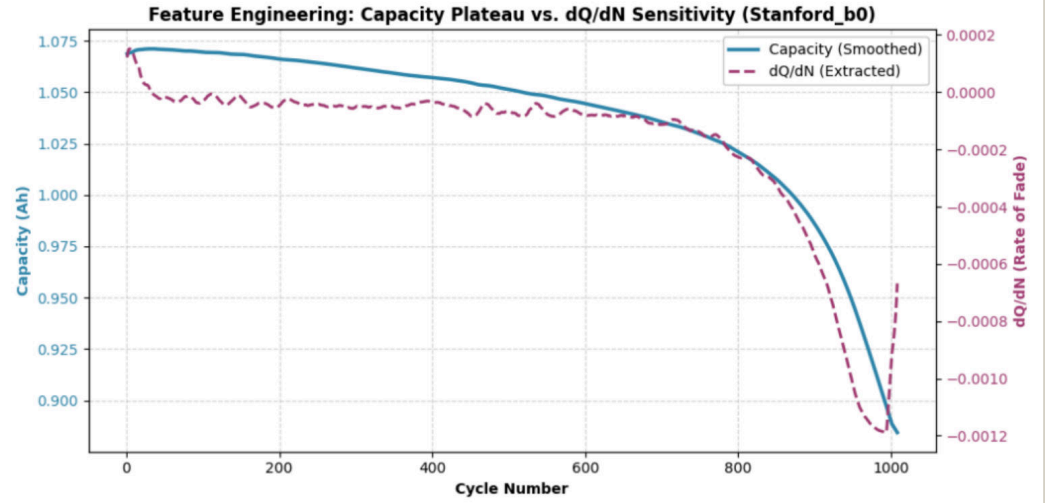


Figure 2 – Raw Stanford Capacity vs. Extracted dQ/dN

FEATURE ENGINEERING — CAPACITY PLATEAU VS. EXTRACTED DQ/DN SIGNAL (STANFORD DATASET)

build_lstm_model.py

```python
def build_lstm_model(input_shape):
    model = Sequential([
        LSTM(64, return_sequences=True, input_shape=input_shape),
        Dropout(0.2),
        LSTM(32, return_sequences=False),
        Dropout(0.2),
        Dense(1)
    ])
    model.compile(optimizer='adam', loss='mean_squared_error')
    return model
```

THE MODEL — TWO-LAYER STACKED LSTM (64 → 32) WITH DROPOUT, ADAM OPTIMISER AND MSE LOSS

02 THIRD-YEAR PROJECT · GROUP OF 6 · 2025

Vibration Analysis — High-Speed-Rail *Suspension with ML Anomaly Detection*

Designed a semi-active suspension concept for a high-speed train — air springs with variable stiffness, a hydraulic actuator and secondary damper — modelled on quarter-car physics. Paired it with an LSTM autoencoder that reconstructs vibration signals and flags anomalies via reconstruction error, triggering an automated speed-response interlock when a fault is confirmed.

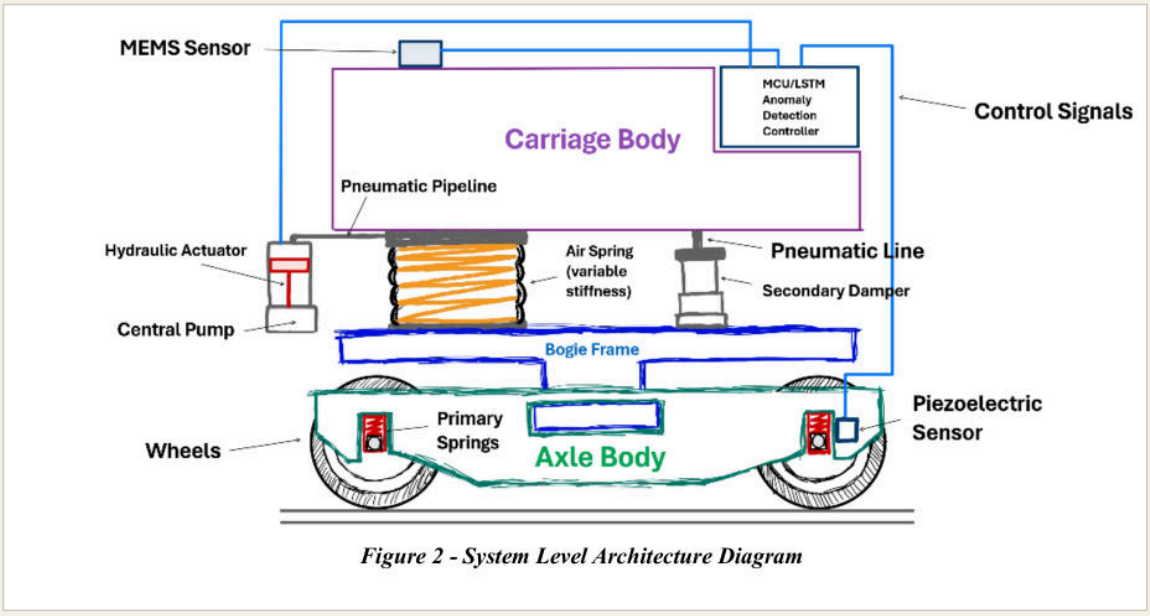
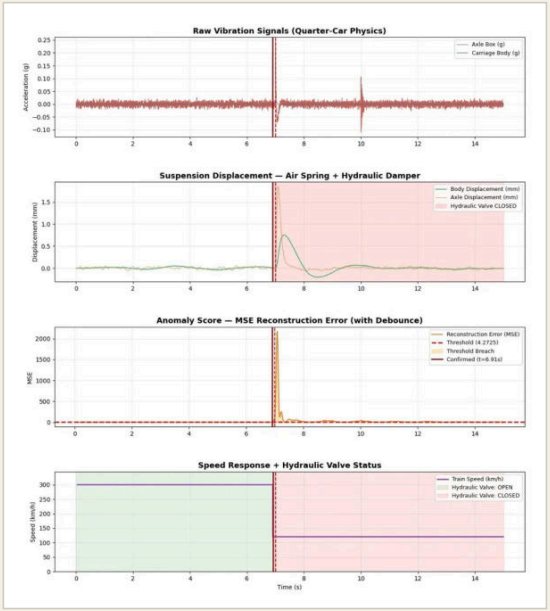


Figure 2 - System Level Architecture Diagram

SYSTEM-LEVEL ARCHITECTURE — AIR SPRING, HYDRAULIC ACTUATOR, SENSING & MCU/LSTM CONTROLLER

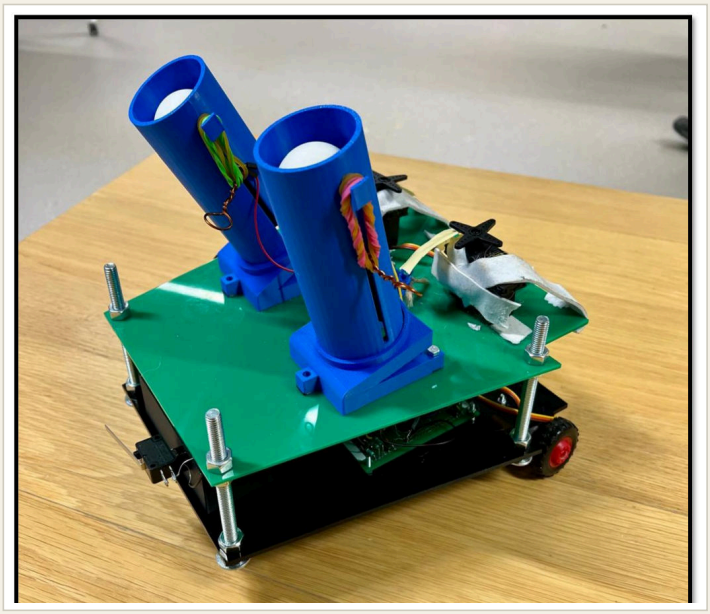


QUARTER-CAR PHYSICS — DISPLACEMENT AND HYDRAULIC VALVE RESPONSE

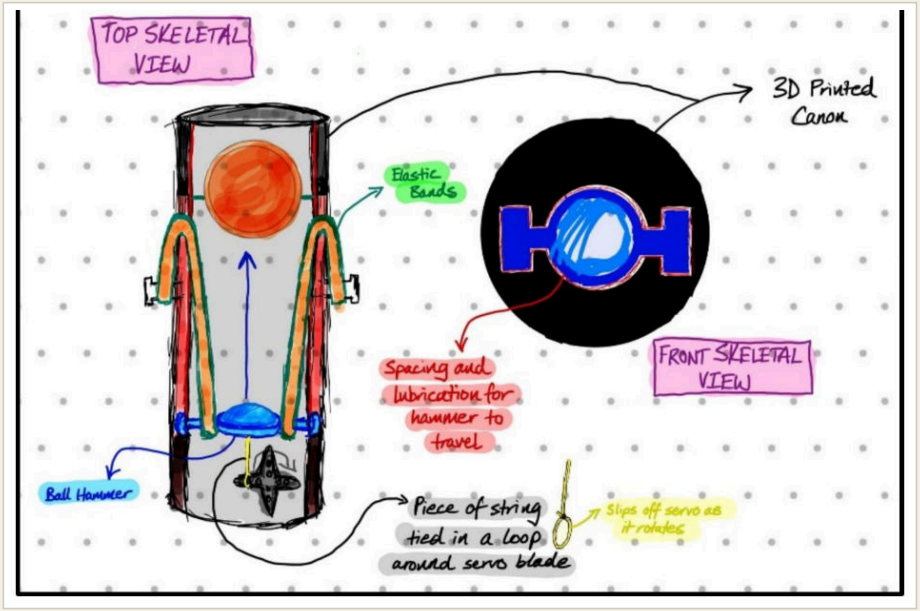
03 SECOND-YEAR PROJECT · TEAM · 2024

Line-Following Robot with a *Dual-Barrel Ping-Pong Cannon*

This was a group project. The team built an autonomous microcontroller-driven robot that follows a line using LDR sensing and fires ping-pong balls at a specified target after travelling a set distance. A dual-barrel mechanism — 3D-printed cannon, elastic-band ball hammer and servo-released string trigger — launches both balls at different heights.



THE BUILT ROBOT — TWIN 3D-PRINTED BARRELS ON THE DRIVE CHASSIS

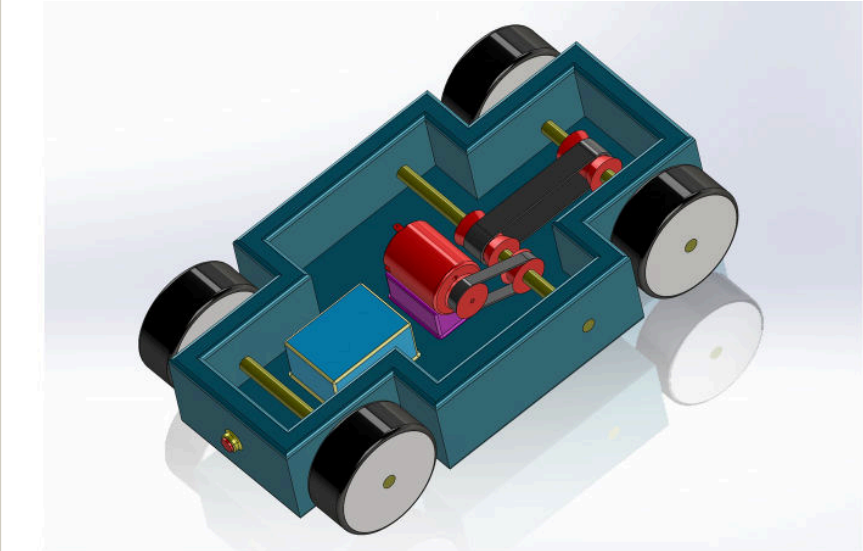


CANNON MECHANISM — BALL HAMMER, ELASTIC BANDS AND SERVO-RELEASED STRING TRIGGER

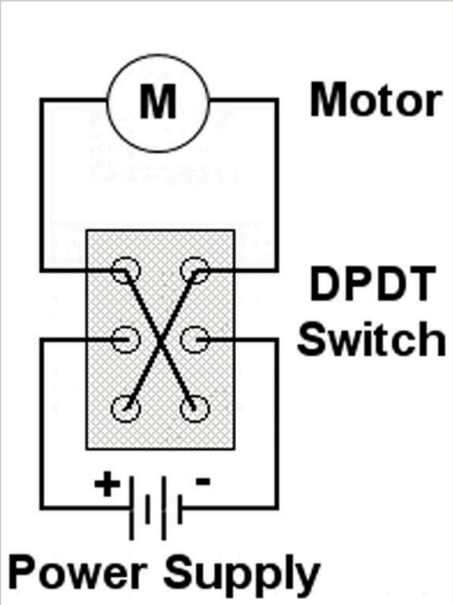
04 FIRST-YEAR PROJECT · IMECHE · 2023

IMechE Autonomous *Wall-Strike & Return Robot Car*

A group project for the IMechE challenge. The team designed and built a robot car programmed to drive into a wall and reverse precisely back into its original starting position — an exercise in deterministic control, distance calibration and mechanical reliability. A DPDT switch reverses the motor direction; the chassis and drivetrain were designed in SolidWorks.



SOLIDWORKS CHASSIS & DRIVETRAIN ASSEMBLY

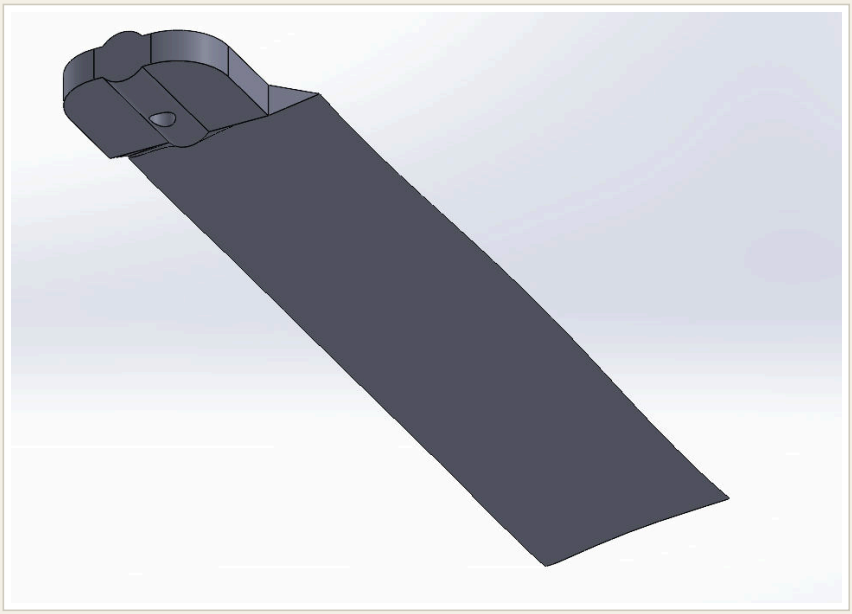


DPDT REVERSING CIRCUIT — MOTOR & POWER SUPPLY

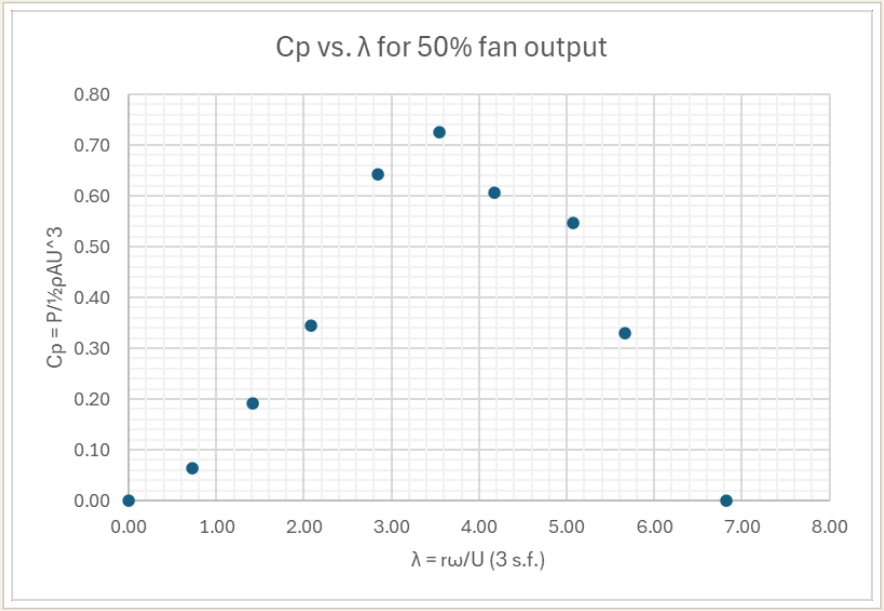
05 FIRST-YEAR PROJECT · GROUP · 2023

Scalable Miniature *Wind-Turbine Aerofoil*

Designed, modelled and wind-tunnel tested a scalable aerofoil blade for maximum power-extraction efficiency. Used JavaFoil and SolidWorks for geometry, then measured current, voltage and power across fan output levels to compute the coefficient of performance and tip-speed ratio.



SOLIDWORKS BLADE GEOMETRY



COEFFICIENT OF PERFORMANCE VS. TIP-SPEED RATIO